Modern Control Design

UAVs

MEAer - Autumn Semestre – 2022/2023

Rita Cunha / J.R.Azinheira

Modern Control Design

→ 1st part based on

Modern Control Design, by Pedro Batista, Feb2018

- State space models
- Control and observation
- Optimal control

→ Example based on

Linear Systems Theory, by J. Hespanha, Princeton Univ. Press, 2009

Linear state space model

- Dynamic model in the time domain:

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t) \end{cases}$$

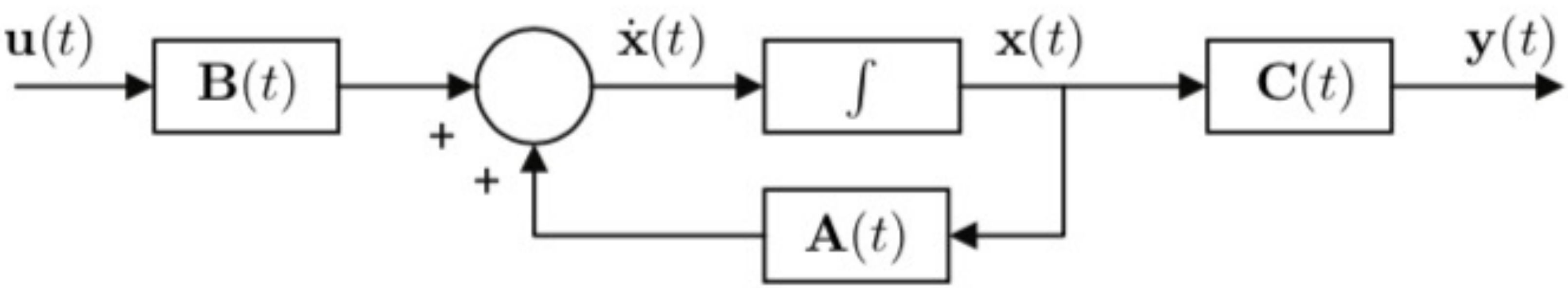
The term $\mathbf{D}(t)\mathbf{u}(t)$ corresponds to a direct effect of the input on the output. That is not possible in general, hence we will always consider $\mathbf{D}(t) = \mathbf{0}$.

A special case occurs when the system matrices are constant, yielding the so-called linear time invariant (LTI) system:

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \end{cases}$$

Linear state space model

- Expressed as a Block diagram
  - Input u(t)
  - State x(t)
  - Output y(t)



TÉCNICO LISBOA

Linear state space model

- Conversion from state-space to transfer function matrix for LTI systems

1. Take the Laplace transform of both sides considering zero initial conditions:

$$\begin{cases} s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s) \\ \mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) \end{cases}$$

2. Determine $\mathbf{Y}(s)$ as a function of $\mathbf{U}(s)$:

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) = \mathbf{C}\left(s\mathbf{I} - \mathbf{A}\right)^{-1}\mathbf{B}\mathbf{U}(s)$$

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \end{cases} \Leftrightarrow \quad \mathbf{G}(s) = \mathbf{C}\left(s\mathbf{I} - \mathbf{A}\right)^{-1}\mathbf{B}$$

Linear state space model

- Solution for LTI systems

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \end{cases}$$

$$\begin{cases} \mathbf{x}(t) = \mathbf{e}^{\mathbf{A}(t-t_0)}\mathbf{x}_0 + \int_{t_0}^{t} \mathbf{e}^{\mathbf{A}(t-\sigma)}\mathbf{B}\mathbf{u}(\sigma)\, d\sigma \\ \mathbf{y}(t) = \mathbf{C}\mathbf{e}^{\mathbf{A}(t-t_0)}\mathbf{x}_0 + \int_{t_0}^{t} \mathbf{C}\mathbf{e}^{\mathbf{A}(t-\sigma)}\mathbf{B}\mathbf{u}(\sigma)\, d\sigma \end{cases}$$

Linear state space models

- Controllability

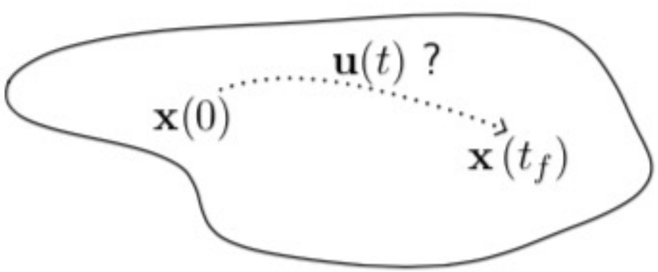
Given the linear state equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t)$$

with initial condition $\mathbf{x}(0) = 0$ and given an arbitrary state $\mathbf{x}_f$, is it possible to make

$$\mathbf{x}(t_f) = \mathbf{x}_f$$

by appropriate choice of $\mathbf{u}(t)$?



The answer depends on the pair of matrices $\mathbf{A}(t)$ and $\mathbf{B}(t)$.

Linear state space models

- Controllability

Definition of controllability

The linear state equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t)$$

is called *controllable on* $[t_0, t_f]$ if, given any initial state $\mathbf{x}(0) = \mathbf{x}_0$, there exists a continuous input signal $\mathbf{u}(t)$ such that the corresponding solution satisfies $\mathbf{x}(t_f) = \mathbf{0}$.

Linear state space models

- Controllability for LTI system

Theorem

The LTI system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

is controllable if and only if the controllability matrix

$$\mathcal{C} := \begin{bmatrix} \mathbf{B} & \mathbf{A}\mathbf{B} & \mathbf{A}^2\mathbf{B} & \dots & \mathbf{A}^{n-1}\mathbf{B} \end{bmatrix}$$

has rank equal to the number of states of the system, i.e.,

$$\operatorname{rank} \mathcal{C} = n, \ \mathbf{x} \in \mathbb{R}^n.$$

Linear state space models

- Controllability for LTI system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

- Controllable system:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= u \end{aligned}$$

$$\mathcal{C} := [\mathbf{B}\ \mathbf{AB}] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Uncontrollable system:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \dot{x}_1 &= x_2 + u \\ \dot{x}_2 &= 0 \end{aligned}$$

$$\mathcal{C} := [\mathbf{B}\ \mathbf{AB}] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

Linear state space models

- Observability

Definition of observability

The linear state equation

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) \\ \mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) \end{cases}$$

is called *observable on* $[t_0, t_f]$ if any initial state $\mathbf{x}(0) = \mathbf{x}_0$ is uniquely determined by the corresponding response $\mathbf{y}(t)$ for $t \in [t_0, t_f]$.

Linear state space models

- Observability

Question: Does anything change if we consider

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) \end{cases}$$

with a known input $\mathbf{u}(t)$, $t \in [t_0, t_f]$?

No!

When the input is known, we can always subtract from the output the component that depends on the input, and thus compute the output corresponding to the zero-input case.

Conclusion: For LTV systems, the observability is independent of the input. In general, that is not the case for nonlinear systems, which makes the analysis much more intricate...

Linear state space models

- Observability for LTI system

Theorem

The LTI system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

is observable if and only if the observability matrix

$$\mathcal{O} := \begin{bmatrix} \mathbf{C} \\ \mathbf{C}\mathbf{A} \\ \vdots \\ \mathbf{C}\mathbf{A}^{n-1} \end{bmatrix}$$

has rank equal to the number of states of the system, i.e.,

$$\operatorname{rank} \mathcal{C} = n, \ \mathbf{x} \in \mathbb{R}^n.$$

Linear state space models

- Observability for LTI system

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \end{cases}$$

- Observable system:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 1 & 0 \end{bmatrix} \qquad \begin{matrix} \dot{x}_1 = x_2 \\ \dot{x}_2 = 0 \end{matrix} \qquad y = x_1$$

$$\mathcal{O} := \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- Unobservable system:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 0 & 1 \end{bmatrix} \qquad \begin{matrix} \dot{x}_1 = x_2 \\ \dot{x}_2 = 0 \end{matrix} \qquad y = x_2$$

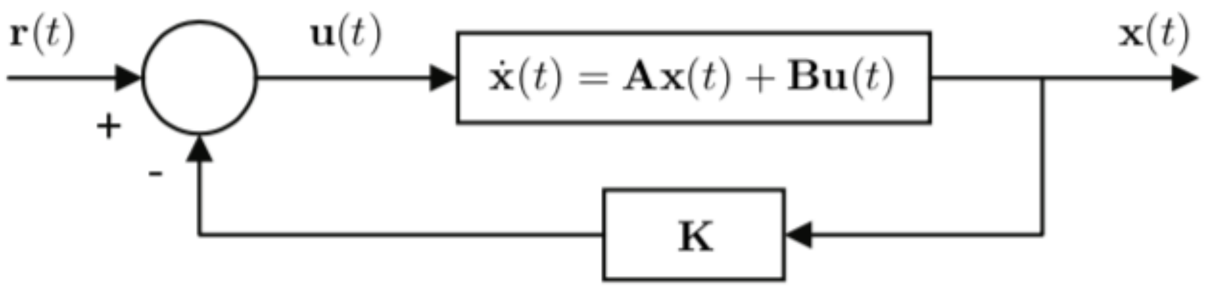
$$\mathcal{O} := \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Linear state space models

- State feedback for LTI system

Is it possible to control the "behavior" of a system using linear state feedback?

Let us focus on LTI systems only, for now... What is linear state feedback?



Linear state feedback:

$$\mathbf{u}(t) = \mathbf{r}(t) - \mathbf{K}\mathbf{x}(t)$$

Linear state space model

- State feedback for LTI system: pole placement

Is it possible to assign the poles of a LTI system using linear state feedback?

Theorem: Pole placement for LTI systems

Suppose that the LTI system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

is controllable. Given any real, monic polynomial $p(\lambda)$ of degree-n, there exists a constant state feedback gain $\mathbf{K}$ such that

$$\det\left(\lambda\mathbf{I} - \mathbf{A} + \mathbf{B}\mathbf{K}\right) = p(\lambda).$$

Linear state space model

- State Observer for LTI system

How do we implement linear state feedback when we don't have access to the entire state of the system?

State observers

Problem: Given the dynamic system

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t) \\ \mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) \end{cases},$$

design a state observer that yields a *"good"* estimate of $\mathbf{x}(t)$.

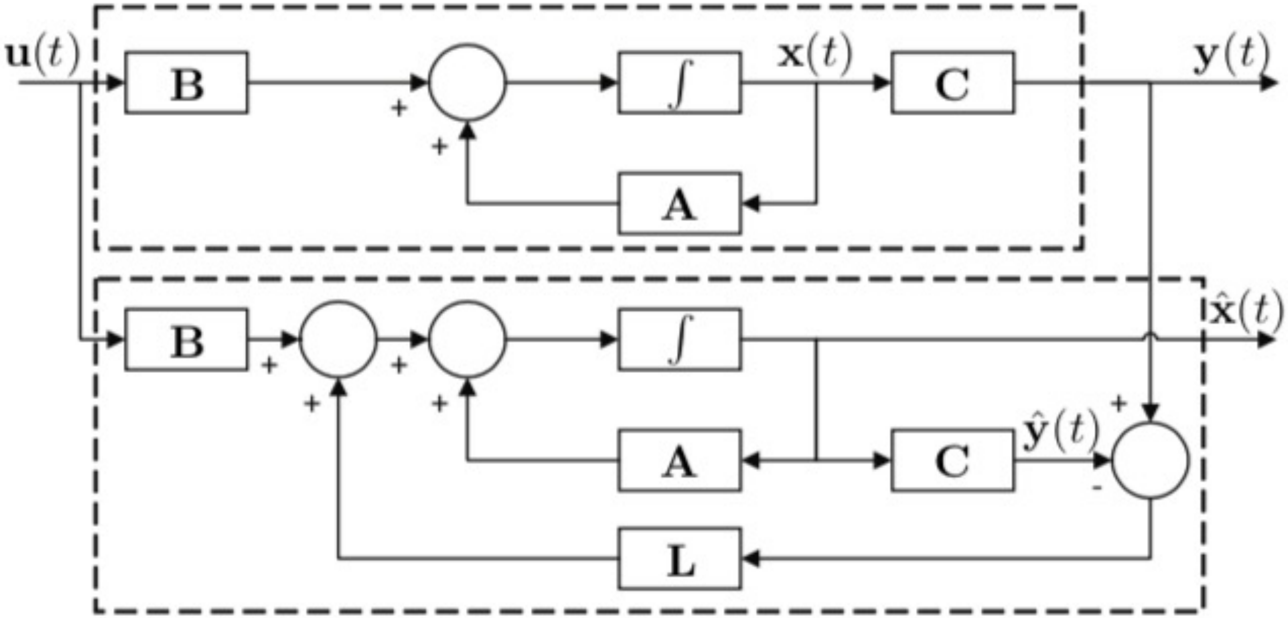
What is a *"good"* estimate of $\mathbf{x}(t)$?

Linear state space model

- State Observer for LTI system: Luenberger observer

Idea: Replica of the system with feedback of the output estimation error

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}(t)\hat{\mathbf{x}}(t) + \mathbf{B}(t)\mathbf{u}(t) + \mathbf{L}(t)\left[\mathbf{y}(t) - \mathbf{C}(t)\hat{\mathbf{x}}(t)\right]$$



Linear state space model

- State Observer for LTI system: Luenberger observer

Observer estimation:

$$\tilde{\mathbf{x}}(t) := \mathbf{x}(t) - \hat{\mathbf{x}}(t)$$

Observer estimation error dynamics:

$$\begin{aligned}
\dot{\tilde{\mathbf{x}}}(t) &= \dot{\mathbf{x}}(t) - \dot{\hat{\mathbf{x}}}(t) \\
&= \mathbf{A}(t)\mathbf{x}(t) - \mathbf{A}(t)\hat{\mathbf{x}}(t) - \mathbf{L}(t)\left[\mathbf{y}(t) - \mathbf{C}(t)\hat{\mathbf{x}}(t)\right] \\
&= \mathbf{A}(t)\tilde{\mathbf{x}}(t) - \mathbf{L}(t)\left[\mathbf{C}(t)\mathbf{x}(t) - \mathbf{C}(t)\hat{\mathbf{x}}(t)\right] \\
&= \left[\mathbf{A}(t) - \mathbf{L}(t)\mathbf{C}(t)\right]\tilde{\mathbf{x}}(t)
\end{aligned}$$

Question: How does one design the observer gain $\mathbf{L}(t)$?

Linear state space model

- State Observer for LTI system: Duality

For linear systems, the control and estimation problems are **dual!**

This is very clear in the LTI case...

The eigenvalues of

$$\mathbf{A} - \mathbf{LC}$$

are the same of

$$(\mathbf{A} - \mathbf{LC})^T = \mathbf{A}^T - \mathbf{C}^T\mathbf{L}^T.$$

Duality

$$\mathbf{A} \longleftrightarrow \mathbf{A}^T$$
$$\mathbf{B} \longleftrightarrow \mathbf{C}^T$$
$$\mathbf{K} \longleftrightarrow \mathbf{L}^T$$

Linear state space model

- State Observer and State feedback

Linear state feedback

+

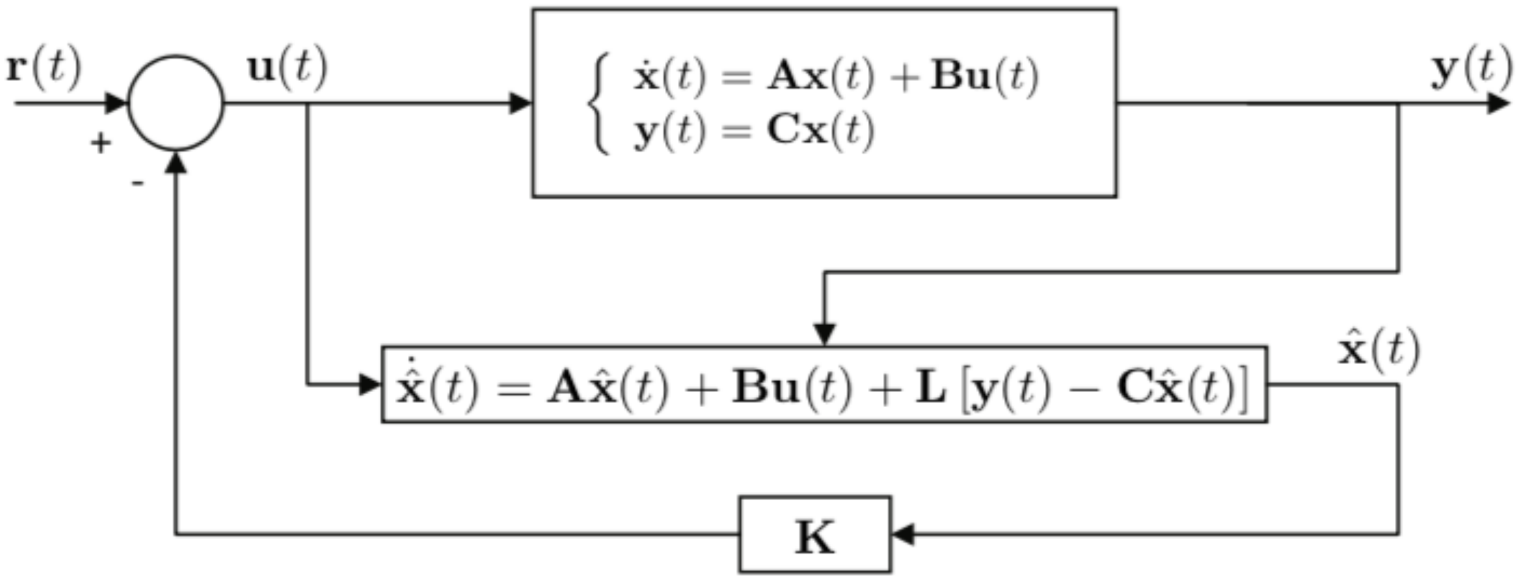
State observers

Does it work?

Can they be designed independently?

Linear state space model

- State Observer and State feedback



Linear state space model

- State Observer and State feedback

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) - \mathbf{B}\mathbf{K}\hat{\mathbf{x}}(t) + \mathbf{B}\mathbf{r}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \\ \dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) - \mathbf{B}\mathbf{K}\hat{\mathbf{x}}(t) + \mathbf{B}\mathbf{r}(t) + \mathbf{L}\left[\mathbf{y}(t) - \mathbf{C}\hat{\mathbf{x}}(t)\right] \end{cases}$$

Recall that, for the observer, we shape the dynamics of the estimation error $\tilde{\mathbf{x}}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t)$.

$$\begin{cases} \dot{\mathbf{x}}(t) = (\mathbf{A} - \mathbf{B}\mathbf{K})\,\mathbf{x}(t) + \mathbf{B}\mathbf{K}\tilde{\mathbf{x}}(t) + \mathbf{B}\mathbf{r}(t) \\ \dot{\tilde{\mathbf{x}}}(t) = (\mathbf{A} - \mathbf{L}\mathbf{C})\,\tilde{\mathbf{x}}(t) \\ \mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) \end{cases}$$

Linear state space model

- State Observer and State feedback for LTI: separation principle

$$\begin{cases} \begin{bmatrix} \dot{\mathbf{x}}(t) \\ \dot{\tilde{\mathbf{x}}}(t) \end{bmatrix} = \begin{bmatrix} \mathbf{A}-\mathbf{BK} & \mathbf{BK} \\ \mathbf{0} & \mathbf{A}-\mathbf{LC} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \tilde{\mathbf{x}}(t) \end{bmatrix} + \begin{bmatrix} \mathbf{B} \\ \mathbf{0} \end{bmatrix} \mathbf{r}(t) \\ \mathbf{y}(t) = \begin{bmatrix} \mathbf{C} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}(t) \\ \tilde{\mathbf{x}}(t) \end{bmatrix} \end{cases}$$

Theorem

The eigenvalues of

$$\begin{bmatrix} \mathbf{A}-\mathbf{BK} & \mathbf{BK} \\ \mathbf{0} & \mathbf{A}-\mathbf{LC} \end{bmatrix}$$

are the union of the eigenvalues of $\mathbf{A}-\mathbf{BK}$ and $\mathbf{A}-\mathbf{LC}$.

Conclusion: The eigenvalues of the controller and the observer may be assigned independently.

Linear state space model

- State Observer and State feedback for LTI: separation principle

  - The separation principle concerns the design of controllers and observers for LTI systems by pole assignement and basically states that these processes can be carried out independently.
  - Separation theorem: for linear systems, a much stronger property can be shown, which concerns the design of **optimal** controllers and observers, which can be carried out, again, independently! More on this later...
  - For nonlinear systems there are no general results.

Linear state space system: optimal control

- Linear Quadratic Regulator

Consider the LTI system

$$\begin{cases} \dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \\ \mathbf{x}(0) = \mathbf{x}_0 \end{cases} .$$

- We know how to *"shape"* the closed-loop dynamics by pole placement.
- Is it possible to do better, to optimize some kind of cost functional?

Linear state space model

- Linear Quadratic Regulator

LQR problem

Given the LTI state equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t),$$

consider the LQR optimization criterion

$$J := \int_0^{\infty} \left(\mathbf{x}^T(t)\,\mathbf{Q}\mathbf{x}(t) + \mathbf{u}^T(t)\,\mathbf{R}\mathbf{u}(t)\right) dt$$

where $\mathbf{Q}$ is positive semi-definite and $\mathbf{R}$ is positive definite.
Problem: What is the control law that minimizes the LQR criterion?

Linear state space model

- Linear Quadratic Regulator

Theorem

Suppose that the pair $(\mathbf{A}, \mathbf{B})$ is stabilizable and the pair $(\mathbf{A}, \mathbf{G})$ is detectable, with $\mathbf{Q} = \mathbf{G}^T\mathbf{G}$. Then:

- there exists a unique, positive definite solution $\mathbf{P}$ of the algebraic Riccati equation

$$\mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} + \mathbf{Q} - \mathbf{P}\mathbf{B}\mathbf{R}^{-1}\mathbf{B}^T\mathbf{P} = \mathbf{0};$$

- The feedback law

$$\mathbf{u}(t) := -\mathbf{K}\mathbf{x}(t), \; \mathbf{K} := \mathbf{R}^{-1}\mathbf{B}^T\mathbf{P}$$

  minimizes the LQR criterion and $J = \mathbf{x}^T(0)\mathbf{P}\mathbf{x}(0)$; and
- The eigenvalues of $\mathbf{A} - \mathbf{B}\mathbf{K}$ all have negative real part.

Linear state space model

- Linear Quadratic Regulator:

→ Relaxation of the controllability and observability conditions

It is sufficient that the pair (A, B) is stabilizable and the pair (A, G) is detectable.

- Stabilizability: (A, B) system is stabilizable if all the uncontrollable modes are stable.
- Detectability: (A, G) system is detectable if all the unobservable modes are stable.

Linear state space model

- Linear Quadratic Regulator

Trade-off between performance and control cost:

- If $\mathbf{Q} \gg \mathbf{R}$, the easiest way to minimize J is to use a lot of control resulting in a small controlled state.
- If $\mathbf{Q} \ll \mathbf{R}$, the easiest way to minimize J is to use little control, resulting in a large controlled state.

Bryson's rule:

$$\mathbf{Q}_{ii} = \frac{1}{\text{maximum acceptable value of } x_i^2(t)}$$
$$\mathbf{R}_{ii} = \frac{1}{\text{maximum acceptable value of } u_i^2(t)}$$

Trial and error...

Modern control design

LQR Design Example

LQR Design Example

- Design an LQR controller for the linearized model of an aircraft roll dynamics[1]
  - Gain selection
  - Analyse effect of changing weights in the frequency domain
  - Connection with loop-shaping

[1]Example extracted from
J. Hespanha, Linear Systems Theory, Princeton University Press, 2009.

LQR Design Example

- Linearized aircraft roll dynamics

$$\dot{\phi} = \omega$$

$$\dot{\omega} = 0.8\omega - 20\tau$$

$$\dot{\tau} = -50\tau + 50u$$

  - In state space form

$$x = \begin{bmatrix} \phi \\ \omega \\ \tau \end{bmatrix} \qquad y = \phi \qquad \begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx \end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -0.8 & -20 \\ 0 & 0 & -50 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 0 \\ 50 \end{bmatrix} \qquad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

LQR Design Example

- Optimal LQR Problem: find u(t) that minimizes

$$J = \int_0^{\infty} \left(x(t)^T Q x(t) + \rho u(t)^2 \right) dt$$

  – Choose

  $Q = G^T G \qquad G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \gamma & 0 \end{bmatrix} \qquad \rho > 0,\ \gamma > 0$ Design weights

  such that

$$J = \int_0^{\infty} \left(z(t)^T z(t) + \rho u(t)^2 \right) dt \quad z(t) = Gx(t) = \begin{bmatrix} \phi \\ \gamma \dot{\phi} \end{bmatrix}$$

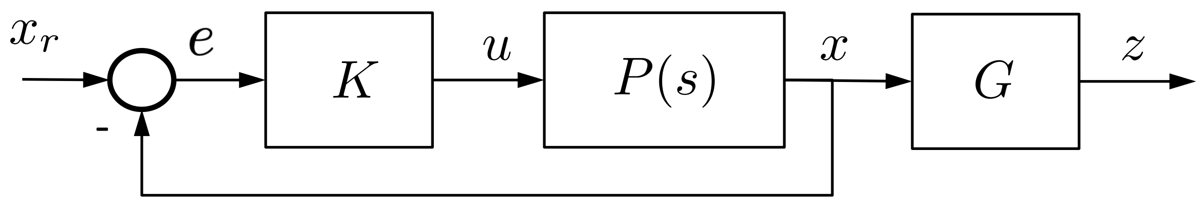
  – LQR solution $u = -Kx$

  – Analyse the effect of changing the weights (ρ, γ) in the frequency domain –> loop shaping

    - Which transfer functions should be considered?

LQR Design Example

- Block diagram and Transfer functions



  - Open-loop transfer function from u to x

    $$P(s) = (sI - A)^{-1}B$$

  - Open-loop negative-feedback transfer function from e to x

    $$L(s) = P(s)K = (sI - A)^{-1}BK$$

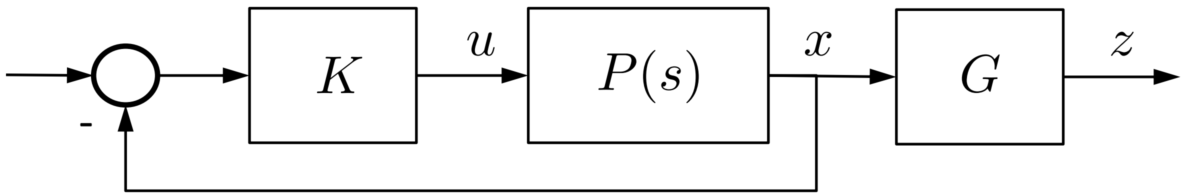
  - Closed-loop transfer function from x_r to x

    $$\dot{x} = (A - BK)x + BKx_r$$

    $$X(s) = (sI - (A - BK))^{-1}BKX_r(s)$$

LQR Design Example

- Block diagram and Transfer functions

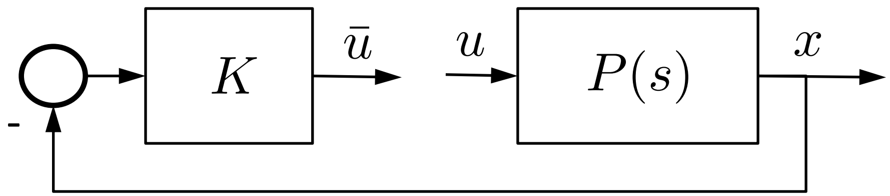


– Open-loop transfer function from u to x

$$P(s) = (sI - A)^{-1}B$$

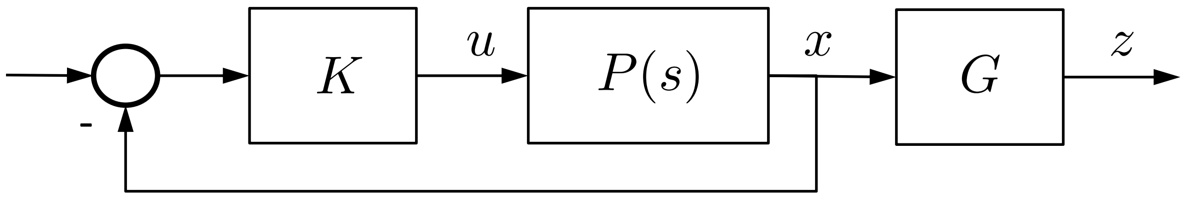
– Open-loop transfer function from u to $-\bar{u}$ (scalar)

$$KP(s) = K(sI - A)^{-1}B = \begin{bmatrix} k_1 & k_2 & k_3 \end{bmatrix} (sI_3 - A)^{-1} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$



LQR Design Example

- Block diagram and Transfer functions



  - Open-loop gain transfer function (scalar)

    $$L(s) = KP(s) = K(sI - A)^{-1}B$$

    can be used as in traditional loop shaping :

    - G_M = ?
    - P_M = ?
    - Noise attenuation (high frequencies)
    - Disturbance rejection, reference tracking (low frequencies)

LQR Design Example

- Kalman's equality (can be derived from ARE)

$$|1 + L(j\omega)|^2 = |1 + KP(j\omega)|^2 = 1 + \frac{\|GP(j\omega)\|^2}{\rho}$$

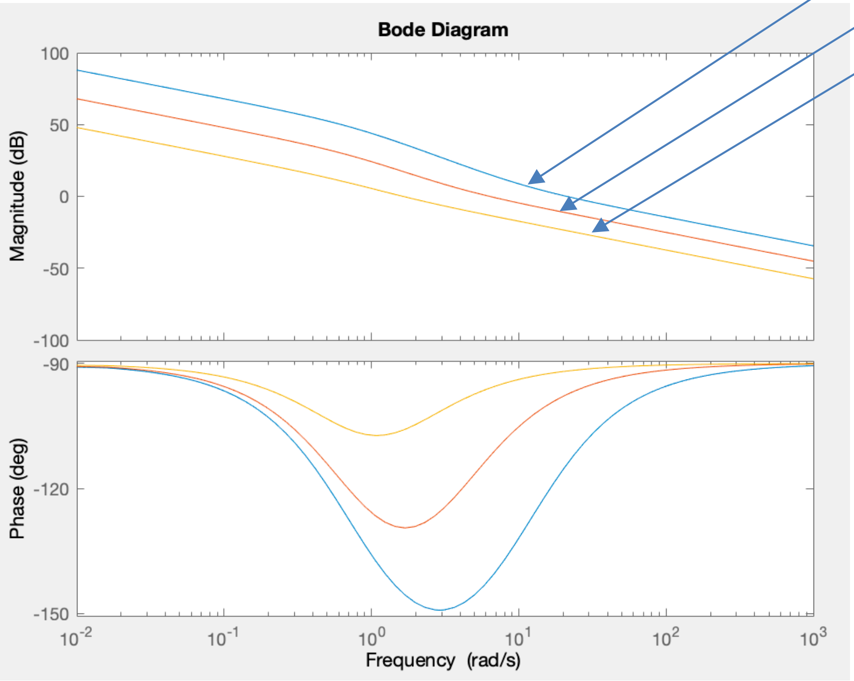
  – With $G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \gamma & 0 \end{bmatrix}$ and $P(s) = \begin{bmatrix} P_1(s) \\ sP_1(s) \\ P_3(s) \end{bmatrix}$ in the low freq.

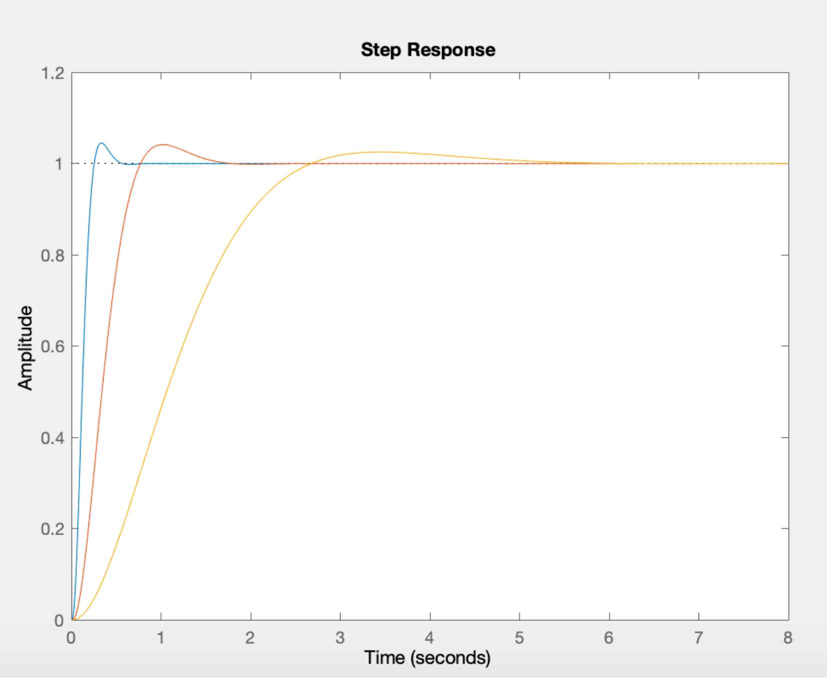
$$|L(j\omega)| \approx \frac{|1 + j\gamma\omega||P_1(j\omega)|}{\sqrt{\rho}}$$

- Effect of changing ρ
  – decreasing ρ moves Bode magnitude plot up
- Effect of changing γ
  – large values of γ lead to a low frequency zero

LQR Design Example

- Example with $\gamma = 0.01 \quad \rho \in \{0.01, 1, 100\}$

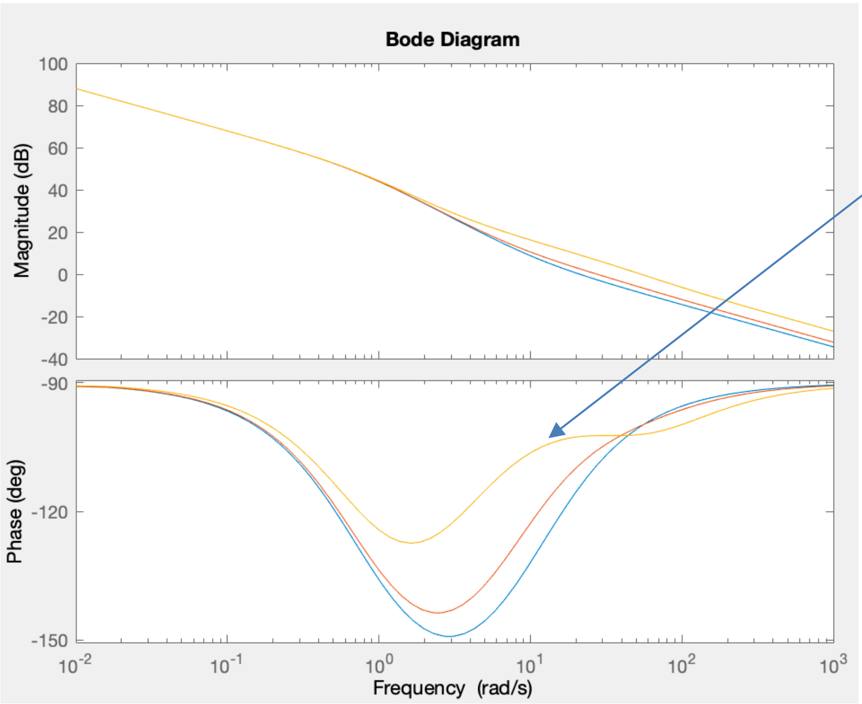


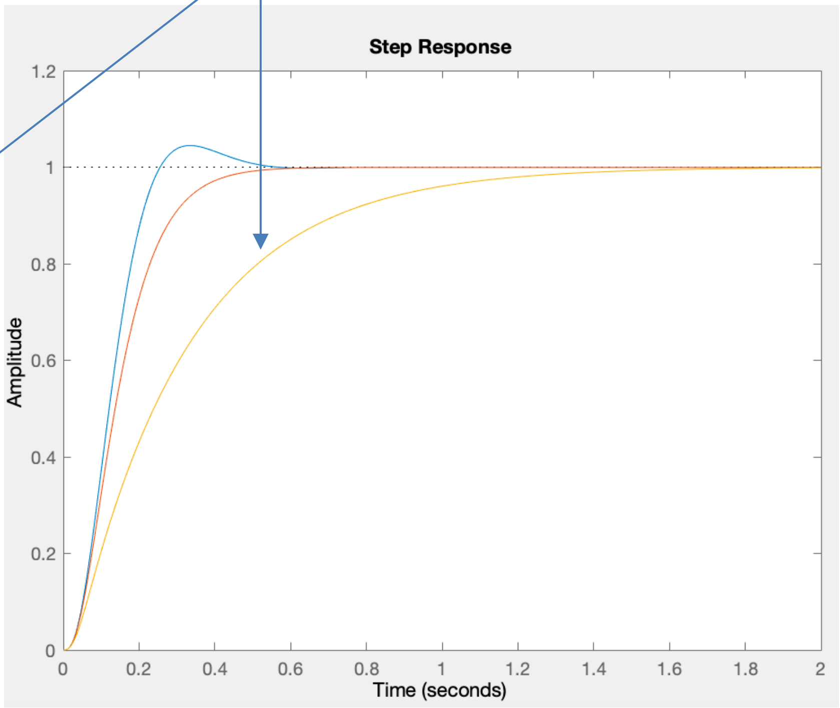


– decreasing ρ moves Bode magnitude plot up
  - faster response, less sensitivity to disturbances
  - reasoning: input becomes "cheaper"

LQR Design Example

- Example with $\rho = 0.01 \quad \gamma \in \{0.01, 0.1, 0.3\}$





  – increasing γ increases the phase margin
    - less overshoot, slower response
    - reasoning: increasing the cost of angular velocity

Luenberger Observer Design Example

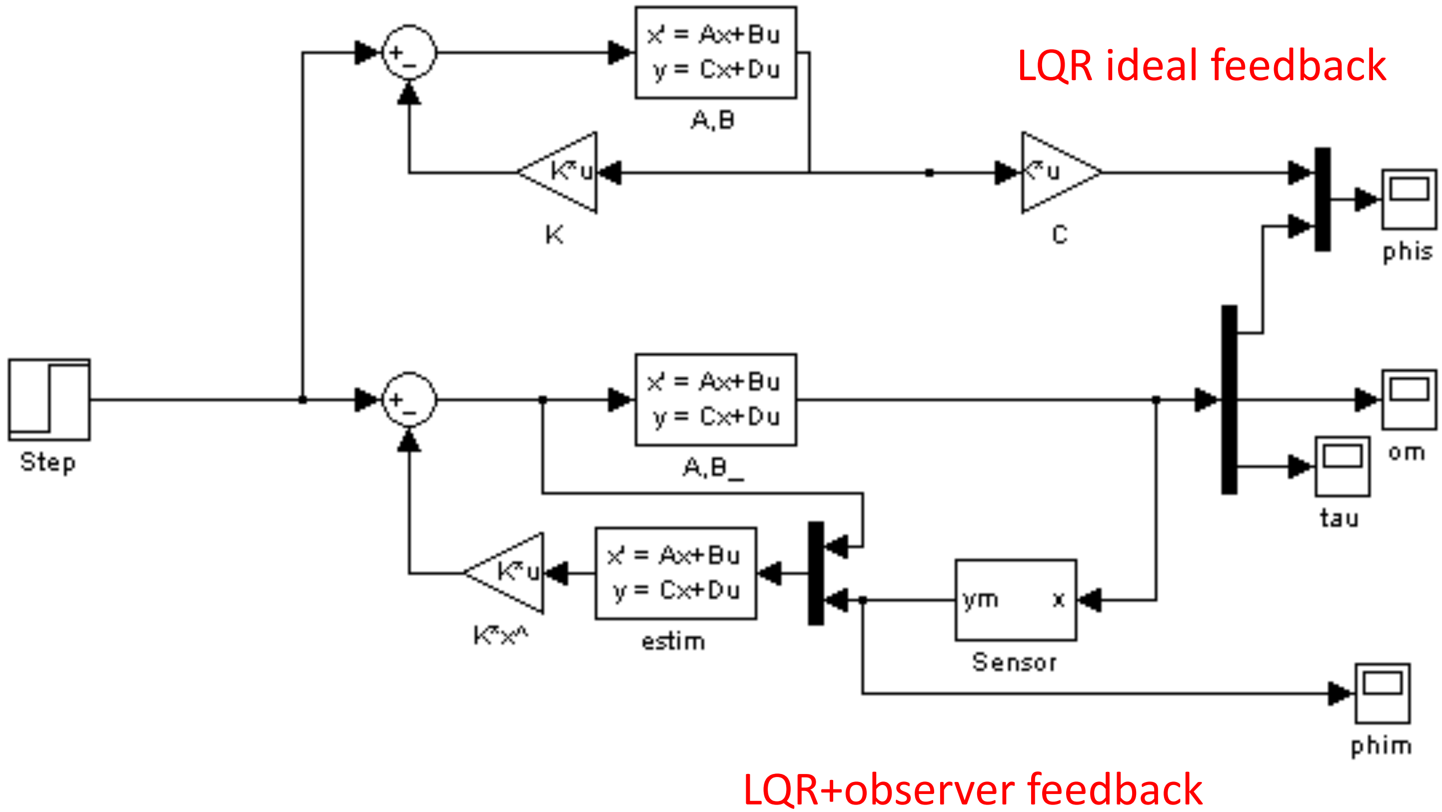
- Using previous model

```
R=1;
G=[1 0 0;0 .3 0]; Q=G'*G
% Q =
%      1.0000         0         0
%           0    0.0900         0
%           0         0         0
K=lqr(A,B,Q,R)
% K =    -1.0000    -0.4105    0.1526
damp(A-B*K)
%            Eigenvalue                 Damping      Freq. (rad/s)
%   -4.40e+000 + 9.05e-001i        9.79e-001        4.49e+000
%   -4.40e+000 - 9.05e-001i        9.79e-001        4.49e+000
%   -4.96e+001                     1.00e+000        4.96e+001


pCL=eig(A-B*K);
L=place(A',C',pCL)' %choose same poles
% L =
%      7.6275
%     29.0897
%     37.9793
esR=estim(ss(A,B,C,0),L,1,1); %build observer
```

Luenberger Observer Design Example

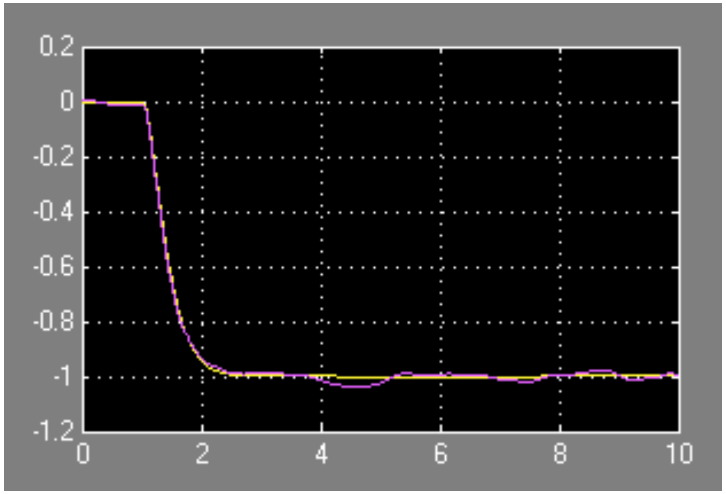
- Simulink implementation

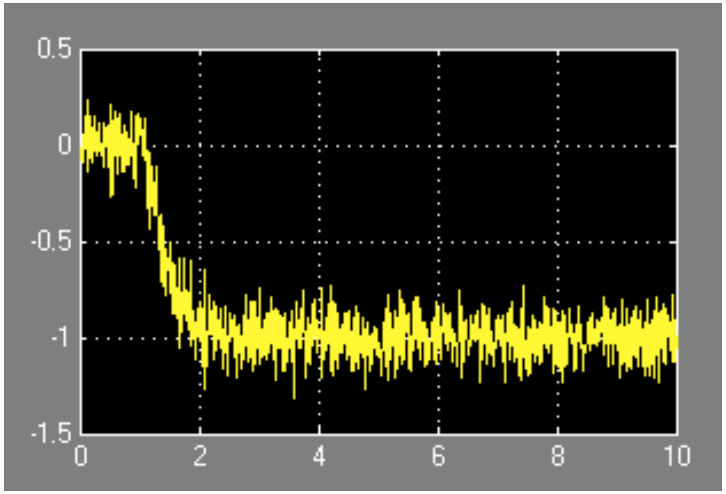


Luenberger Observer Design Example

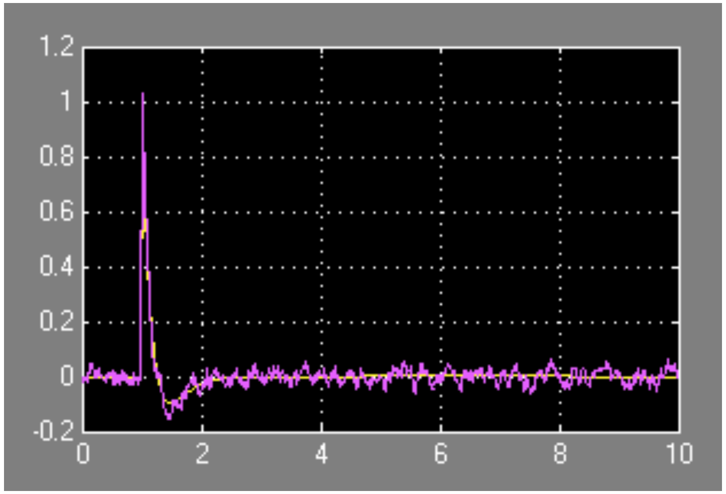
- Simulink implementation

measurement





outputs



Control inputs